

Weekly Plan & Progress Report (Week10)

SUN RUI A0330128R

JIANG YIBAI A0332309L

NUS-ISS

Doxa Holdings | Technology Department

Week 10 Report (2026-05-01 to 2026-05-07)

1. Milestones Achieved

1. **Completed the end-to-end A2A-AML-POC solution design** — finalized the business context, input/output definitions, risk grading criteria, and final report structure, creating a unified target for implementation.
2. **Established the core tech stack and architecture path** — completed key technology selections across runtime, agent orchestration, LLM abstraction, HTTP services, storage, RAG retrieval, and observability, with rationale and maturity assessment for each.
3. **Finalized the three-agent collaboration model (Host/Market/Transaction)** — defined clear responsibility boundaries, AgentCard skills, A2A message interaction patterns, and the HostAgent's LangGraph-based iterative flow: dispatch, evaluate, and re-dispatch.
4. **Completed the system engineering blueprint** — delivered the full directory structure, key file responsibility map, SQLite data model (tasks/messages/artifacts/sessions), and data lifecycle flow.
5. **Completed the end-to-end interaction and demo flow design** — documented the sequence from user input to a final CRITICAL risk report, including evidence-chain generation logic and a CLI streaming demo path.
6. **Documented quick start and initialization procedures** — provided setup steps for environment preparation, mock data generation, embedding build, database initialization, and one-command startup to reduce onboarding and integration effort.

2. Milestones Missed

1. **Did not complete a runnable implementation of the three-agent services**
(design/specification only)
 - Reason: this week prioritized architecture convergence and interface contracts to avoid downstream rework.
 - Resolution: next week, implement the base A2A layer and three-agent service skeleton first, then incrementally complete business logic.
2. **Did not complete real end-to-end integration and performance validation**

- Reason: executable services and data initialization pipeline are not yet implemented, preventing stable integration testing.
 - Resolution: after enabling a minimum runnable path (single case), run SSE chain validation, latency checks, and error-recovery tests.
3. **Did not complete reproducible delivery of mock data and embedding artifacts**
- Reason: script flow is designed but has not entered execution and verification.
 - Resolution: prioritize implementation and hardening of generate_mock_data and build_embeddings scripts, with validation steps and sample outputs.

3. Plan and Effort Adjustments

- **2026-05-01**: shifted this week's goal from "code first" to "freeze architecture and protocol first" to avoid repeated interface churn in multi-agent collaboration. Decision: prioritize complete technical design documentation.
- **2026-05-03**: confirmed POC scope as "3 core agents" instead of implementing all 16 business agents at once, to reduce first-version complexity and validate A2A collaboration value quickly. Decision: focus on the Host/Market/Transaction minimum closed loop.
- **2026-05-05**: adjusted storage strategy to SQLite-first, emphasizing low cost and fast validation in POC stage. Decision: stabilize four core tables (tasks/messages/artifacts/sessions) before evaluating migration to production-grade databases.
- **2026-05-07**: adjusted RAG strategy to "NumPy vector retrieval + lightweight knowledge base" to avoid premature vector database operations overhead. Decision: deliver demonstrable retrieval capability first, then upgrade infrastructure as needed.

4. Forecasted Risks and Mitigations

1. **Risk of mismatch between A2A protocol implementation and SSE streaming details** — may cause unstable inter-agent calls or state-machine failures.
 - Mitigation: implement a unified contract in a2a/types.py and run minimum interoperability tests before layered expansion.
2. **Uncontrolled LLM cost and response latency** — multi-round collaboration may increase both cost and interaction delay.
 - Mitigation: enforce call budgets and timeout policies, validate high-value nodes first, and reduce redundant rounds.
3. **Insufficient mock data quality may reduce demo credibility** — incomplete coverage can produce unstable anomaly-detection conclusions.

- Mitigation: expand samples by anomaly type (price anomaly, related-party transaction, structuring) and add regression test cases.
4. **Execution gap from documentation to implementation** — strong design output but delayed implementation pace may impact upcoming milestones.
- Mitigation: execute a four-step plan next week (base protocol layer -> three-agent skeleton -> single-case integration -> report generation), with verifiable outputs at each step.

Week 11 Plan (2026-05-08 to 2026-05-14)

1. Work Assignment

- **Implement the A2A base protocol layer:** complete minimum viable versions of types, server, client, and registry, supporting /tasks/send and /tasks/sendSubscribe.
- **Build the three-agent service skeleton:** bring up HostAgent, MarketAgent, and TransactionAgent FastAPI services and load AgentCards.
- **Enable a minimum closed-loop core workflow:** connect HostAgent's main chain (parse -> dispatch -> evaluate -> report), covering at least one Tesla high-price transaction case.
- **Complete data and storage initialization:** implement mock data generation, embedding build, and SQLite initialization scripts for one-command reproducibility.
- **Integration and validation:** complete CLI streaming demo, key log tracing, and failure-path validation (timeout/empty results/call failure).

2. Expected Milestones

1. **Three-agent services can start independently and call each other:** local runtime is stable, and both A2A requests and SSE subscriptions are observable.
2. **First end-to-end investigation case is fully runnable:** anomalous transaction input produces a structured report including risk level, anomaly type, evidence chain, and recommended action.
3. **Data and scripts are reproducible:** standardized steps from environment initialization to demo execution allow same-day onboarding and reproduction by new team members.
4. **POC technical risks are exposed early:** deliver an integration issue list with repair priorities to support expansion to more agents.